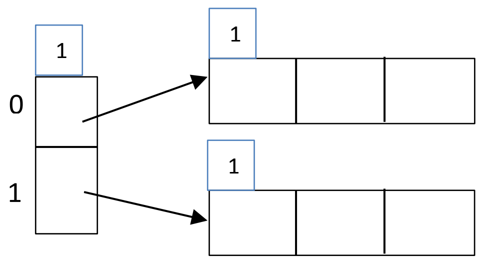
Homework #3 - Solutions

1. Insert the records with the keys **85, 73, 16, 79, 34, 38, 26, 39,** and **29** into a hash table using **extendible hashing.** Use **h(key) = key mod 33** as the hash function. Assume that the global depth is equal to 1 and the hash table is empty at the beginning. Each data page may hold up to 3 records. Show each step of the insertion.

Answer: The initial hash table is given as follows:



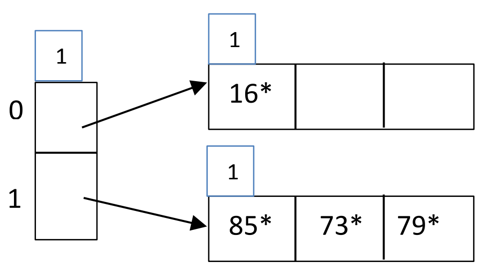
Then, insert 85, 73, 16, 79 as follows:

h(85) = 85 mod 33 = 19 = $(1001\mathbf{1})_2$

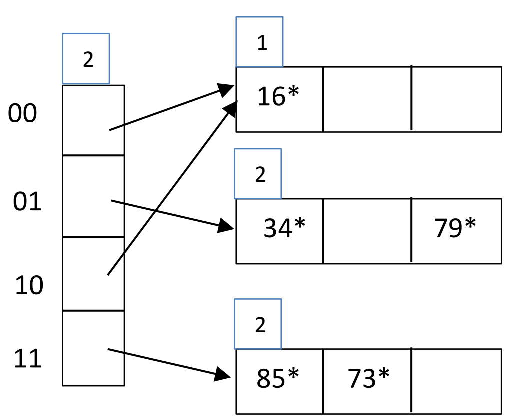
h(73) = 73 mod 33 = 7 = $(11\mathbf{1})_2$

h(16) = 16 mod 33 = 16 = $(1000\mathbf{0})_2$

h(79) = 79 mod 33 = 13 = $(110\mathbf{1})_2$

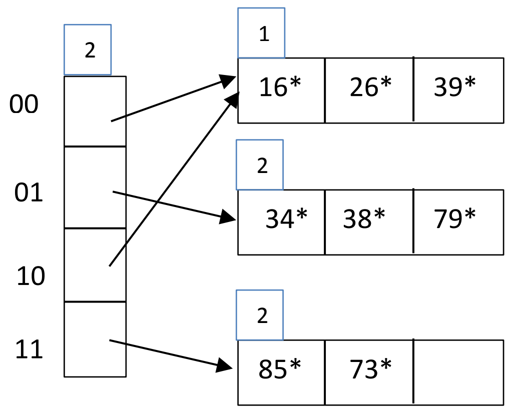


h(34) = 34 mod 33 = 1 = $(\mathbf{1})_2$ To insert 34, we need to double the directory size.



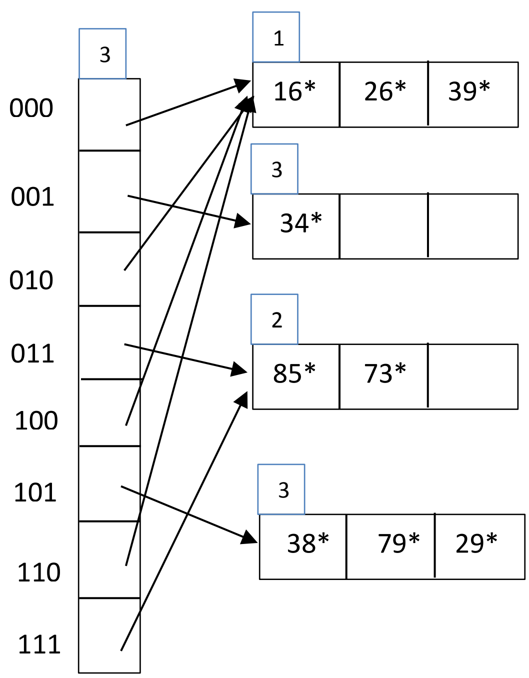
Then insert 38, 26, 39.

h(38) = 38 mod 33 = 5 = $(1\mathbf{01})_2$
h(26) = 26 mod 33 = 26 = $(110\mathbf{10})_2$
h(39) = 39 mod 33 = 6 = $(1\mathbf{10})_2$



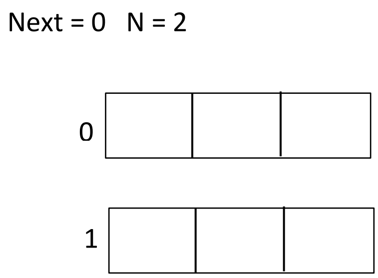
Then insert 29.

h(29) = 29 mod 33 = 29 = $(111\mathbf{01})_2$ To insert 29, we need to double the directory size.

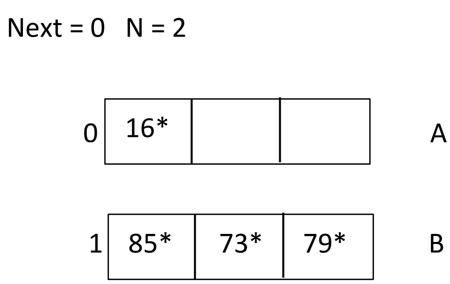


2. Redo problem 1 with linear hashing. Assume that N = 2, and next = 0.

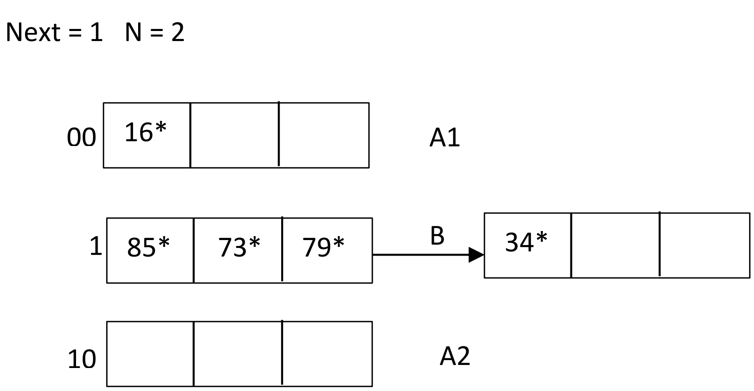
Answer: The initial hash table is given as follows:



Now, insert 85, 73, 16, 79 as follows:
h(85) = 85 mod 33 = 19 = $(10011)_2$
h(73) = 73 mod 33 = 7 = $(111)_2$
h(16) = 16 mod 33 = 16 = $(10000)_2$
h(79) = 79 mod 33 = 13 = $(1101)_2$



h(34) = 34 mod 33 = 1 = $(1)_2$ To insert 34, we need to divide page pointed by next which we call as A into two. Then insert an overflow page to B.



To insert 38, h(38) = 38 mod 33 = 5 = $(101)_2$ we need to divide page pointed by next which we call as B into two. Then redistribute records between B1 and B2, and delete overflow page if it becomes empty. Then update next to 0 and N to 4, since all pages are divided.

Next = 0   N = 4

00	16*			A1
01	34*	38*	79*	B1
10				A2
11	85*	73*		B2

Then insert 26 and 39.

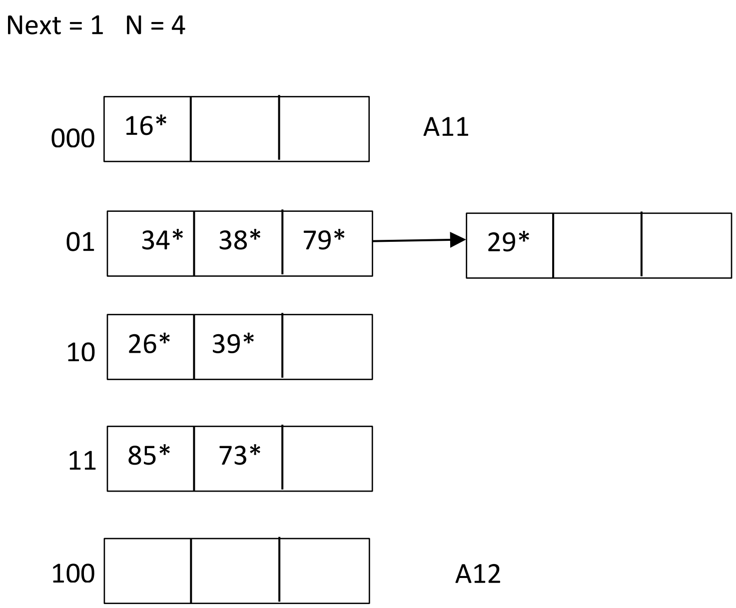
$h(26) = 26 \bmod 33 = 26 = (110\mathbf{10})_2$
$h(39) = 39 \bmod 33 = 6 = (1\mathbf{10})_2$

Next = 0   N = 4

00	16*			A1
01	34*	38*	79*	B1
10	26*	39*		A2
11	85*	73*		B2

Then insert 29.

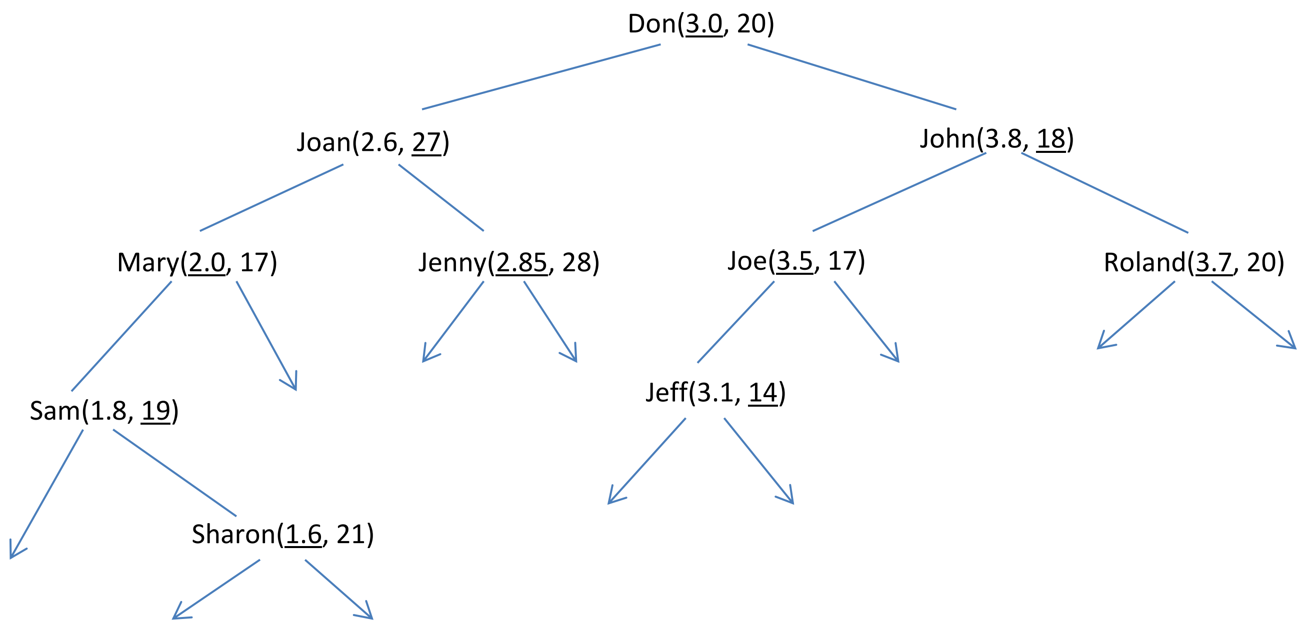
$h(29) = 29 \bmod 33 = 29 = (111\mathbf{01})_2$ To insert 29, we need to divide page pointed by next which we call as A1 into two. Then insert an overflow page to B1.



3. Construct a 2-d tree using the following data with gpa as the first dimension and age as the second dimension etc. Insert the records to the tree starting from the 1st record in the data, use the order given in the data.

Name	gpa	age
Don	3.0	20
John	3.8	18
Joe	3.5	17
Joan	2.6	27
Mary	2.0	17
Jenny	2.85	28
Sam	1.8	19
Roland	3.7	20
Jeff	3.1	14
Sharon	1.6	21

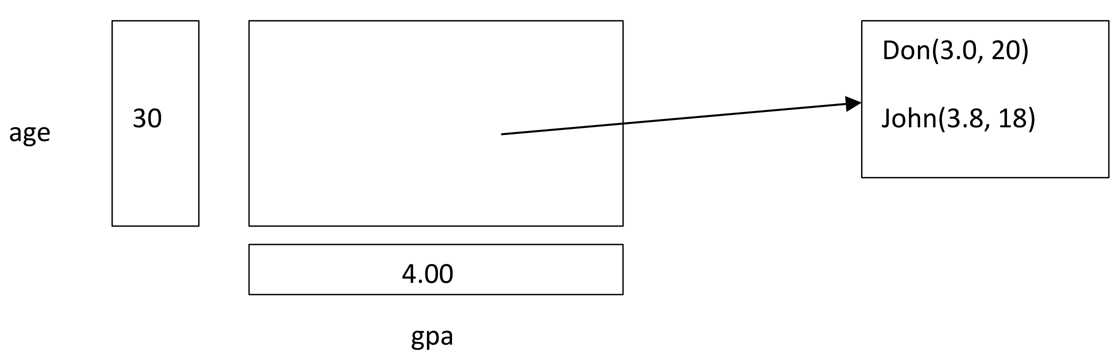
Answer: The 2-d tree developed for the given dataset is shown below:



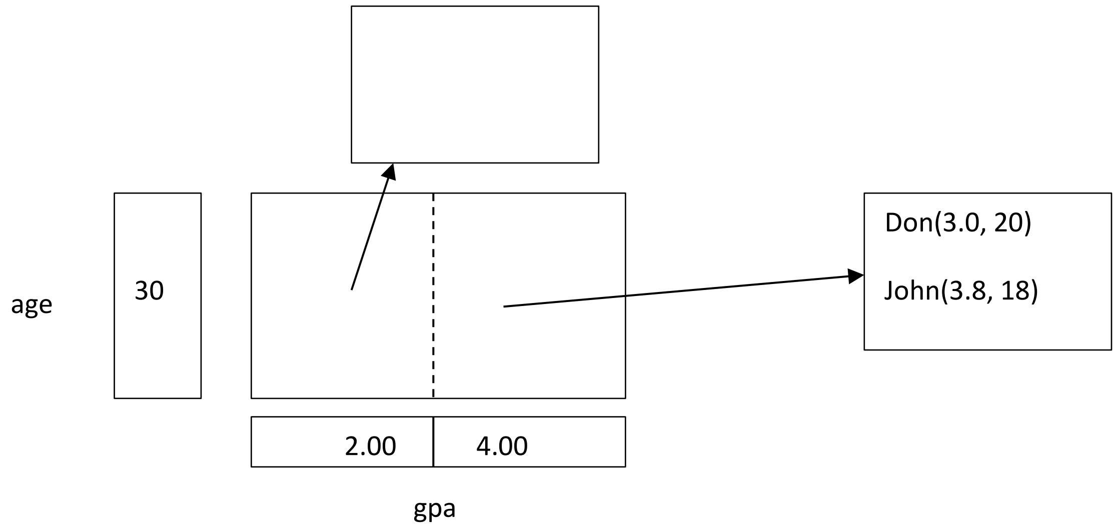
Arrows in the above tree point to the data pages.

4. Construct a grid file using the data given in the 3rd question with gpa as the first dimension to partition and age as the second dimension, etc. Let the gpa dimension range from 0 to 4.00 and the age dimension range from 10 to 30. Assume that each bucket has a capacity of two records.

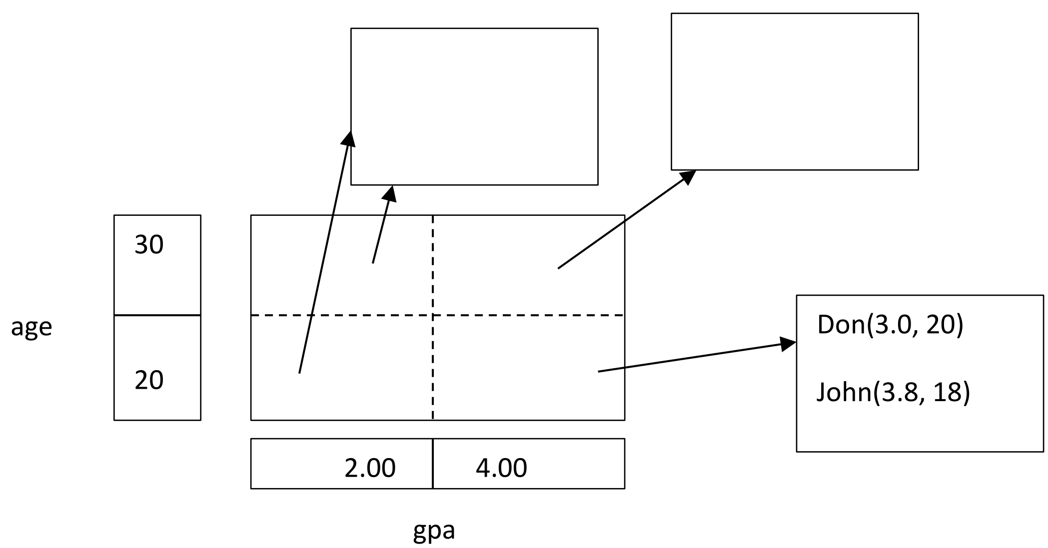
Answer: Initially, we insert the first two records.



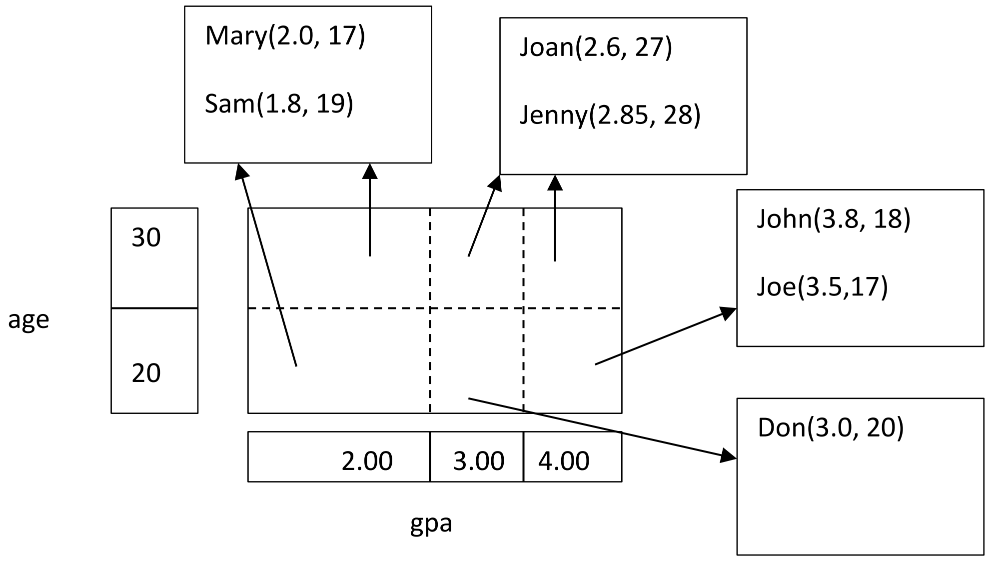
To insert Joe(3.5, 17) we partition the grid from gpa dimension.



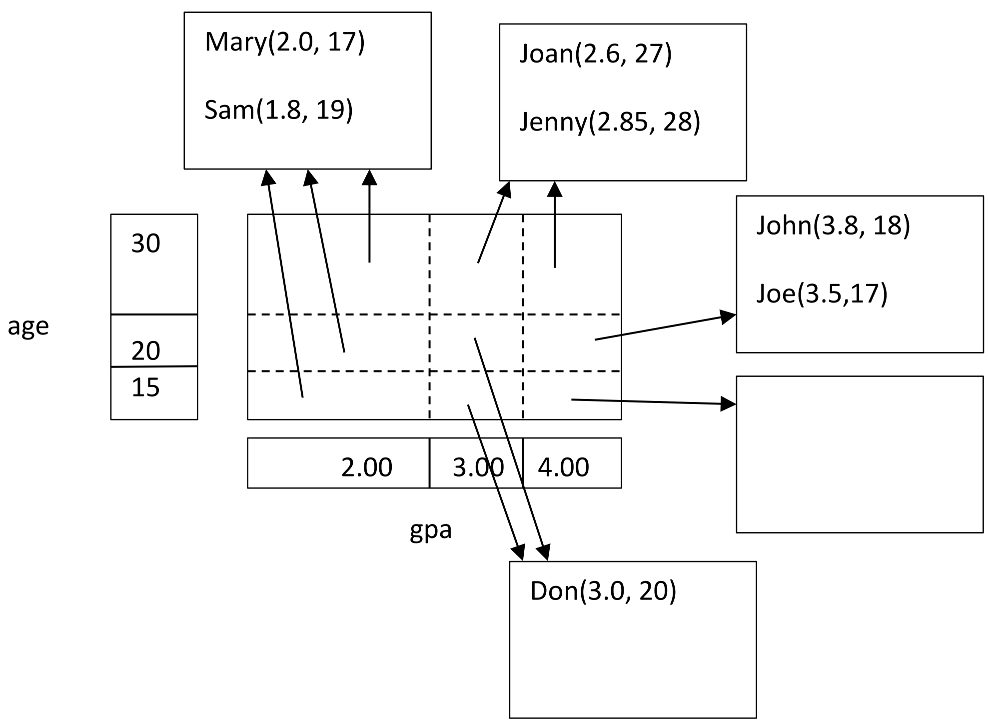
The above partition is not enough to insert Joe(3.5, 17). Now, we partition the grid from age dimension.



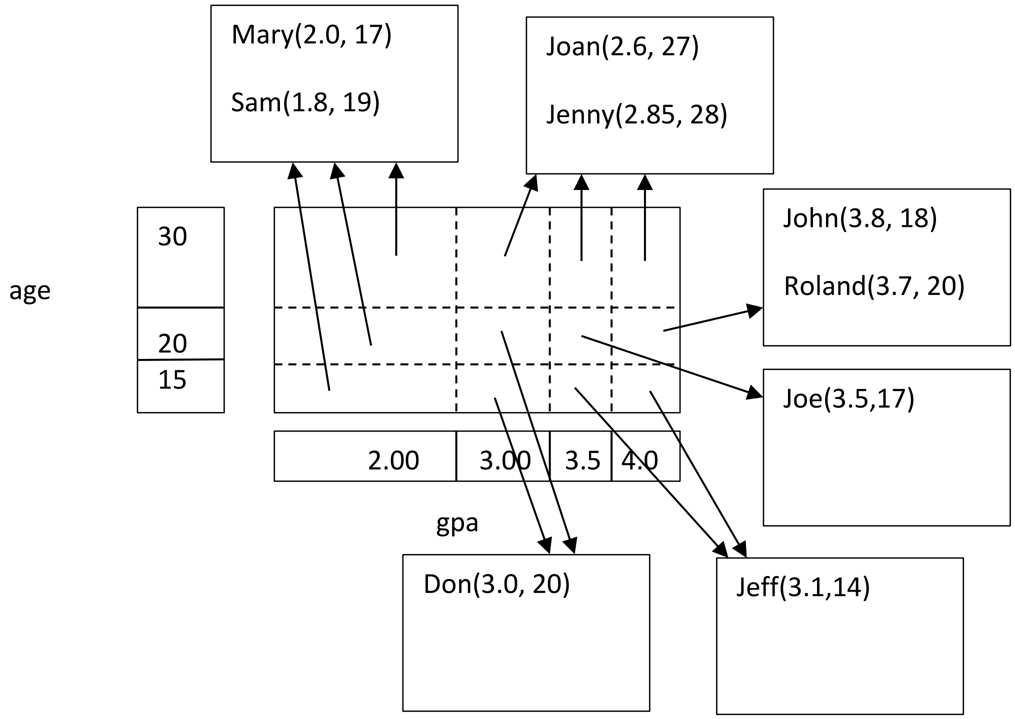
The above partition is also not enough to insert Joe(3.5, 17). Now, we partition the grid from gpa dimension.



To insert Roland(3.7, 20) we partition the grid from age dimension.



The above partitioning is not enough. To insert Roland(3.7, 20) we partition the grid from gpa dimension.



To insert Sharon(1.6, 21) we only divide the page into two, not the grid. The final grid file is shown below: